

Ankit Aggarwal

ankitagg@andrew.cmu.edu | <https://ankitaggarwal0.github.io/> | <https://github.com/AnkitAggarwal0> | (412) 726-4393

Education

Carnegie Mellon University, Master of Science in Robotic Systems Development (MRSD) Expected May 2026
GPA 4.11, Coursework – Systems Engineering, Controls, Autonomy, Motion Planning, Robot Learning, Deep Learning
Manipal Institute of Technology, B. Tech in Mechatronics Engineering (Minor in Robotics) Oct 2020 – Jun 2024
GPA 9.50, Coursework – Kinematics, Dynamics, Controls, Sensors, Electronics, Mechanical Design

Work Experience

Industrial Robotics Intern, Manufacturing Futures Institute, CMU – Pittsburgh, PA May 2025 – Aug 2025

- Deployed ROS-I control on FANUC ArcMate 50iD/7L, replacing manual workflows with automated DXF-based weld path execution
- Engineered remote I/O control of Keyence PLC using Python RPCs for time-synchronized welds, achieving > 90 % repeatability
- Programmed FANUC RoboGuide (KAREL) to stream joint states to ROS, enabling MoveIt-based motion planning and visualization
- Developed diagnostics, sensor integration, and recovery procedures that cut programming/setup time by 90 % (2 h → 10 min)

MITACS Globalink Research Intern (Robotics), Continuum Robotics Laboratory, University of Toronto Jun 2023 – Aug 2023

- Built a 3-DOF additively manufactured tendon-driven continuum robot with custom BLDC actuators and a Nitinol backbone
- Implemented TI Launchpad-based PID tension control with motor encoder feedback, achieving 0.8 mm position accuracy
- Validated system robustness and repeatability across 3 field-deployed prototypes and 50 closed-loop cycles each
- Developed a fully-actuated TDCR endoscope prototype for manual teleoperation and research in constrained environments

Application Engineer Intern – Embedded Systems, STMicroelectronics – India Dec 2021 – Jan 2022

- Built a BLE-controlled omnidirectional guide robot (125 kg payload, 1 m/s) using Mecanum wheels for navigation demonstrations
- Engineered 1 kHz PWM motor actuation via peripheral timers and implemented IMU-based feedback for heading stabilization
- Programmed WB55 BLE wireless module with STM32 (Embedded C) for real-time joystick-driven motion via mobile interface
- Designed 24 V power circuit with buck converters, achieving 90 % efficiency and 2 h battery runtime under full load

Projects

Mars Rover Manipal – Technical Head / Engineer [Project Video](#)

- Led a 40-member team to build a 50 kg Mars Rover prototype for the University Rover Challenge (placed 1st in Asia)
- Achieved 2nd Worldwide (European Rover Challenge 2022) and 3rd Worldwide (International Rover Challenge 2023)
- Designed a 6-DOF robotic arm with a custom 60:1 cycloidal gearbox and 3-DOF wrist, achieving 5 kg payload capacity at 1 m reach
- Implemented inverse kinematic PID control with IMUs for the arm, and manufactured a chassis with a custom 5-bar suspension
- Hands-on fabrication using FDM, CNC, and manual machining; completed certified training in CAD (SolidWorks CSWP), CAM, CAE

Lunar ROADSTER – MRSD Capstone Project, CMU github.com/Lunar-ROADSTER

- Built an Autonomous Moon-working Mechatronic Rover capable of grooming the lunar surface using a bulldozing mechanism
- Developed rover autonomy using convex optimization (GLOP) on terrain point cloud maps for traversable path planning in ROS2
- Fused ZED stereo and IMU data via an extended Kalman Filter for autonomous navigation (< 5 % pose error) and manipulation
- Trained a YOLO v8 model to extract crater geometry using RGB-D data and validated using a point cloud-based gradient map

Object-Centric Dexterous Manipulation for Humanoids – Robot Learning Project, CMU

- Developed a Transformer-based decoder for 6-DOF wrist trajectory prediction from human motion RGB video (ARCTIC dataset)
- Optimized a low-level policy for finger joint control using PPO, achieving an 87.77% completion rate for 11 different objects
- Retargeted the policy from high-DoF Shadow Hands to a Unitree G1 (Inspire hands) in Isaac Sim using geometric mapping and FK
- Migrated the ObjDex hierarchical baseline, originally for fixed-base dual-arms in Isaac Gym, to a humanoid robot in Isaac Sim

Design and Simulation of 3-RPS, 3-RRS, and 3-UPU Parallel Manipulators [Publication](#)

- Developed kinematic/dynamic models of 3-RPS, 3-RRS, and 3-UPU manipulators, modeling joint clearance using six virtual links
- Quantified trajectory deviation using Monte Carlo (66 λ -angles $\times$ 73 actuator settings), showing 18 % error at 0.1 R vs < 5 % ideal
- Validated ~ 12 % workspace reduction under dimensional deviations, setting tolerance guidelines for parallel manipulator design

Skills

Robotics: ROS1/ROS2, MoveIt, RViz, ROS-Industrial, Gazebo, MuJoCo, OMPL, CHOMP, PILZ, Nav2, Dynamics Modeling, URDF/Xacro
Controls and Learning: LQR, MPC, DIRCOL, IPOPT, Nonlinear and Optimal Control, RL, PPO/TRPO, Imitation Learning, CNNs, RNNs
Embedded and Hardware: STM32, PWM, Sensors, Motor Control, Arduino, Power Electronics, 3D Printing, System Integration
Software and Design: C++, Python, Docker, Git, Linux, Embedded C, MATLAB, SolidWorks (CSWP), Fusion 360, Ansys Workbench